Richard Leyton-Romero

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Summary

Controls engineer and PhD researcher specializing in optimal control of nonlinear dynamic systems, with hands-on experience supporting professional motorsport programs.

Education

University of the Witwatersrand – PhD Electrical Engineering 2026

- **Thesis:** The Use of Optimal Control in the Development of Motorsport Performance
  - Research sponsored by General Motors/Cadillac Formula 1 team.

Oxford Brookes University – MSc Motorsport Engineering 2023

- **Dissertation:** The Effects of Race Vehicle Setup on Lap-by-Lap Tire Warm-Up Response

University of South Florida – BSME Mechanical Engineering 2020

Experience

Independent Engineering Consultant, United States 2019 – 2023

Provided engineering support to organizations such as Pirelli Motorsport, Ian Lacy Racing, Ferrari of North America, Riley Technologies, and Multimatic Motorsports.

- Optimized race vehicle setup and stability characteristics through telemetry and driver feedback analysis, improving driver confidence and vehicle consistency.
- Used driver-in-the-loop simulation to identify circuit-specific performance limitations and pre-event setup deficiencies, reducing trackside iteration time.
- Analyzed tire thermal behavior and wear characteristics during race events to optimize tire usage strategy and maximize performance consistency.

Engineering Intern, Comp Cams – Memphis, TN Summer 2019

- Solved a common Chrysler engine failure by identifying critical point of defect and redesigning hydraulic valvetrain lifter to allow for free flow of oil.
- Tuned control parameters for vintage Chevrolet engine retrofitted with modern electronic fuel injection by monitoring and adjusting air:fuel ratios and spark timing.
- Provided powertrain consulting for race teams across various motorsport categories from professional drag racing to boat racing.

Publications

Laptime Driven Optimization of Vehicle Suspension Systems (2026). Leyton Romero, R.D., & Limebeer, D.J.N. *Vehicle System Dynamics*. doi.org/10.1080/00423114.2026.2674200

An Optimal Control Approach to the Generation of Yaw-Moment Diagrams (2025). Limebeer, D. J. N., Leyton Romero, R.D., & Massaro, M. *Vehicle System Dynamics*. doi.org/10.1080/00423114.2025.2471344

Skills

Scientific Computing: MATLAB, Julia, Python, Maple, Mathematica
Motorsport Data Visualization: MoTeC i2, Cosworth Pi, WinTax4